Wrapline

Built on the Zama FHE protocol

WRAPLINE

The Confidential Wrapper Registry, turned into a product.

Ethereum Sepolia + Mainnet • ERC-20 ↔ ERC-7984 • EIP-712 user-decryption

wrapline.vercel.app

THE PROBLEM WITH

the registry today

Without a product layer

FRAGMENTED

Look-alike tokens everywhere

Every team ships against its own mock ERC-20s and wrappers instead of the canonical ones.

01 **Duplicated tokens**

Developers spin up their own testnet ERC-20s and ERC-7984 wrappers rather than reuse the official Zama Wrappers Registry.

02 **Integrations don't compose**

Each app targets a slightly different set of tokens, so confidential assets from one app don't interoperate with the next.

03 **Wallets full of noise**

Users end up holding confidential assets that only *look* like the real thing — the ecosystem fragments.

ONE REGISTRY, *every pair*

Wrapline is the missing product layer on the official registry — reading the on-chain Wrappers Registry as the source of truth, with a local overlay for dev-only pairs.

01 **Browse**

Every canonical ERC-20 ↔ ERC-7984 pair on Sepolia + Mainnet, with live metadata: name, symbol, decimals, addresses.

02 **Wrap & unwrap**

Convert any registry ERC-20 into its confidential twin and back — clean approvals, resumable async unwrap.

03 **Decrypt any ERC-7984**

Reveal your own balance of *any* confidential token via EIP-712 — registry or arbitrary paste-in.

04 **Sepolia faucet**

Claim the official cTokenMocks in one click and try the whole flow immediately.

Hybrid sourcing • on-chain primary + local overlay • deduped, on-chain always wins

LIVE DEMO

END-TO-END *on Sepolia*

Connect a wallet and run the full loop — six steps, no leaving the app.

01 **Browse the registry**

Every canonical ERC-20 ↔ ERC-7984 pair, live metadata, dual-chain.

02 **Claim from the faucet**

Mint an official cTokenMock on Sepolia in one click.

03 **Wrap a cTokenMock**

Approve → encrypt → wrap 1:1 into its confidential twin.

04 **Decrypt the balance**

Reveal the new ERC-7984 balance via EIP-712.

05 **Unwrap back**

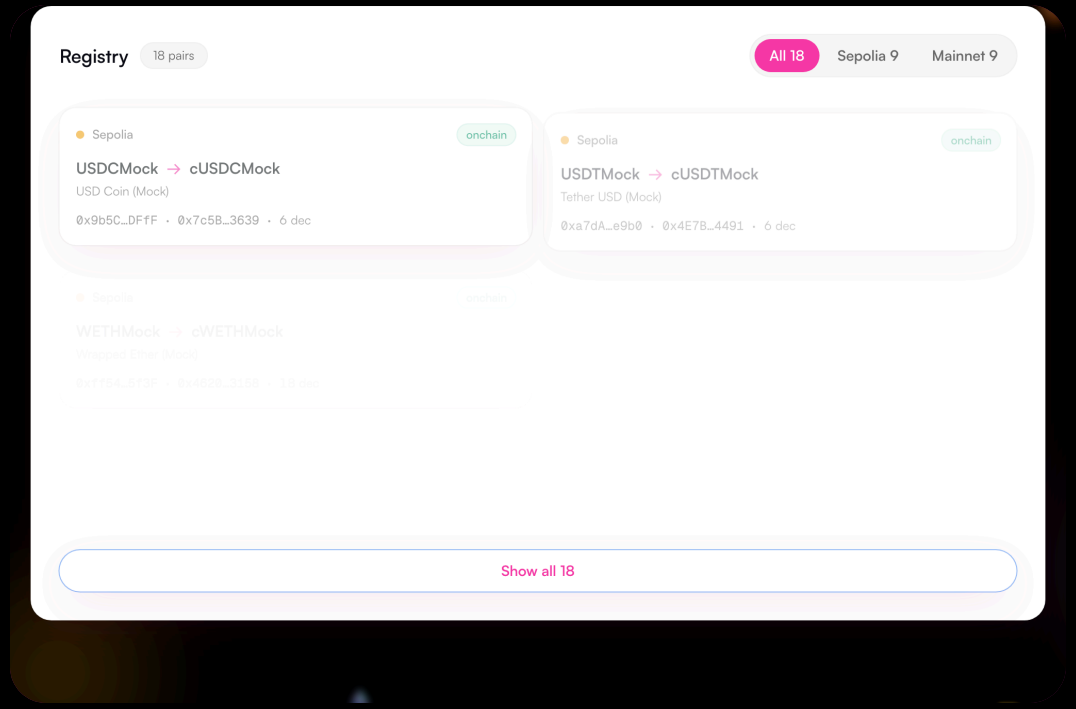
Async request → KMS decrypt → finalize; resumable.

06 **Decrypt any ERC-7984**

Paste any confidential token — even outside the registry.

COVERAGE

BROWSE *the registry*



All 8 official cTokenMocks surfaced from the on-chain Wrappers Registry, plus a dual-chain filter — **Sepolia 9 • Mainnet 9**.

name • symbol • decimals • both addresses

On-chain is the source of truth; a local overlay fills dev-only pairs, deduped.

COVERAGE

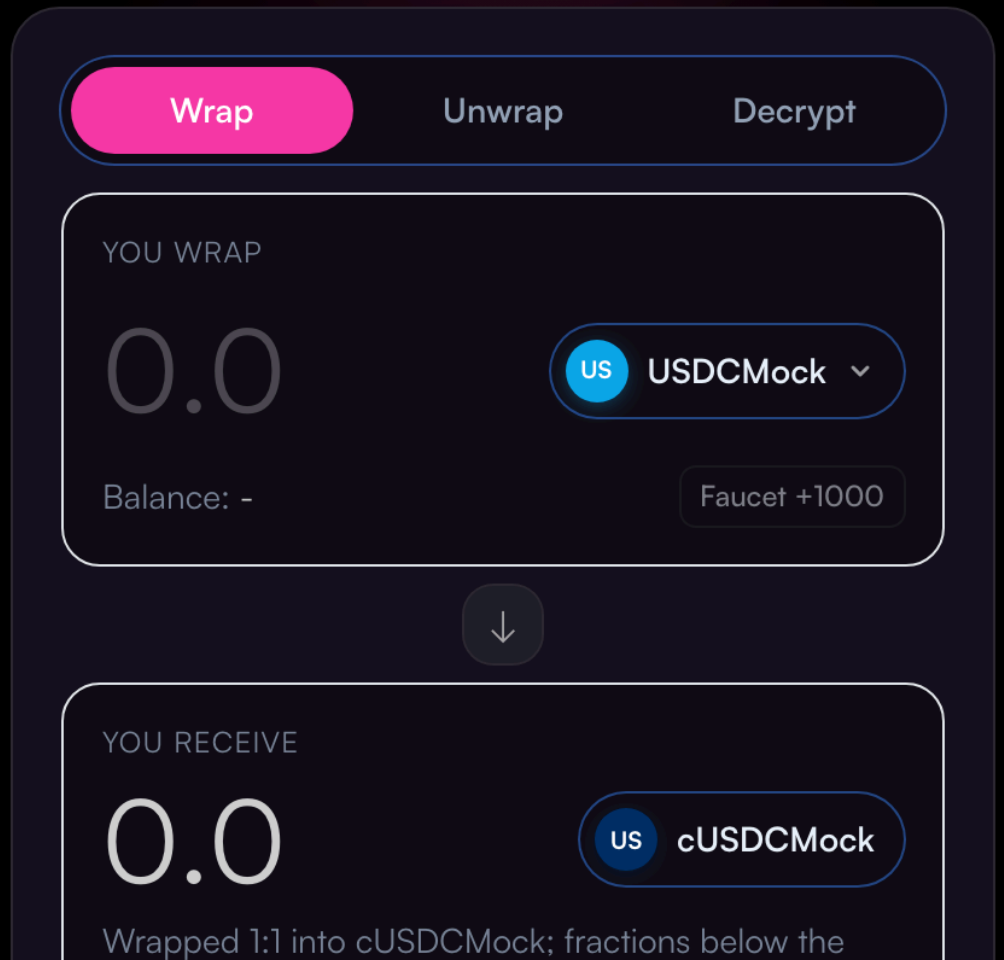
CLAIM *from the faucet*

One click on **Faucet +1000** mints the official cTokenMock ERC-20 straight to your wallet on Sepolia.

mint → balance updates → ready to wrap

Per-address cap aware — the button disables when a claim would exceed the cap.

Wrap, unwrap, and decrypt confidential tokens.



CORRECTNESS

WRAP *a cTokenMock*

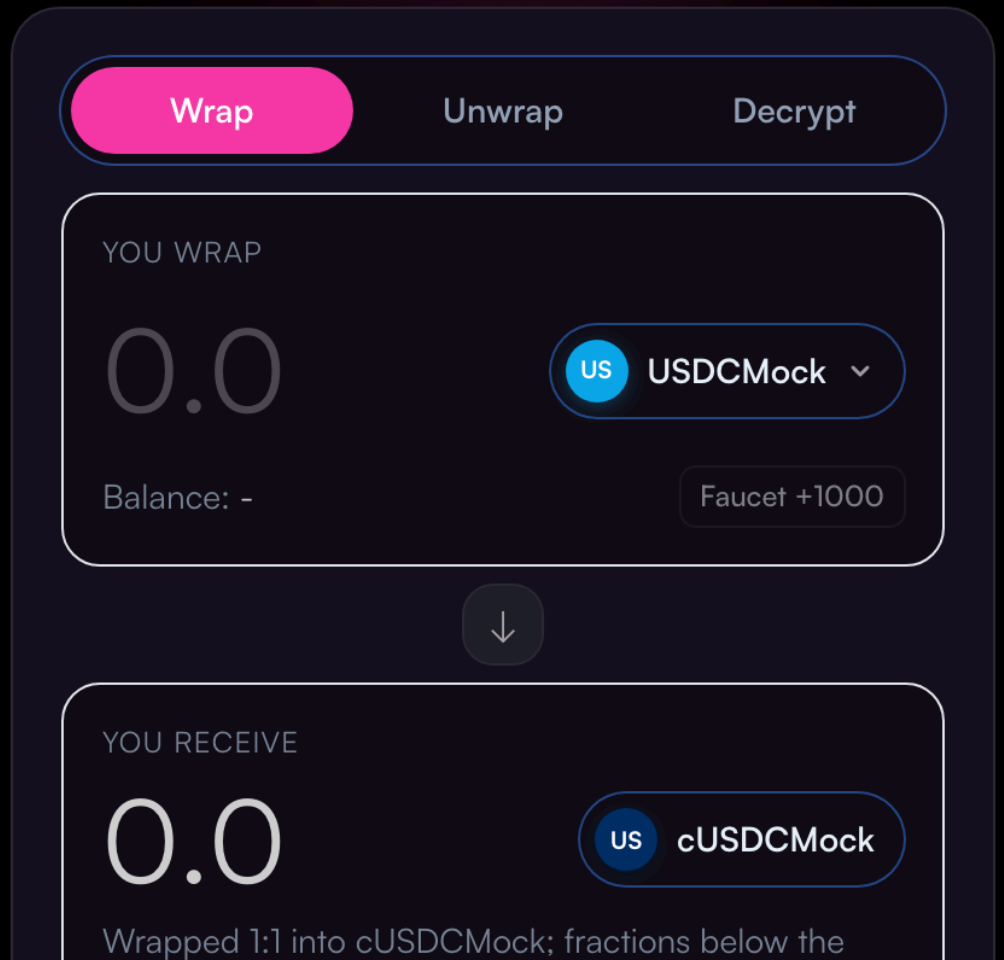
1 **Approve ERC-20 allowance**

2 **FHE-encrypt the amount** client-side

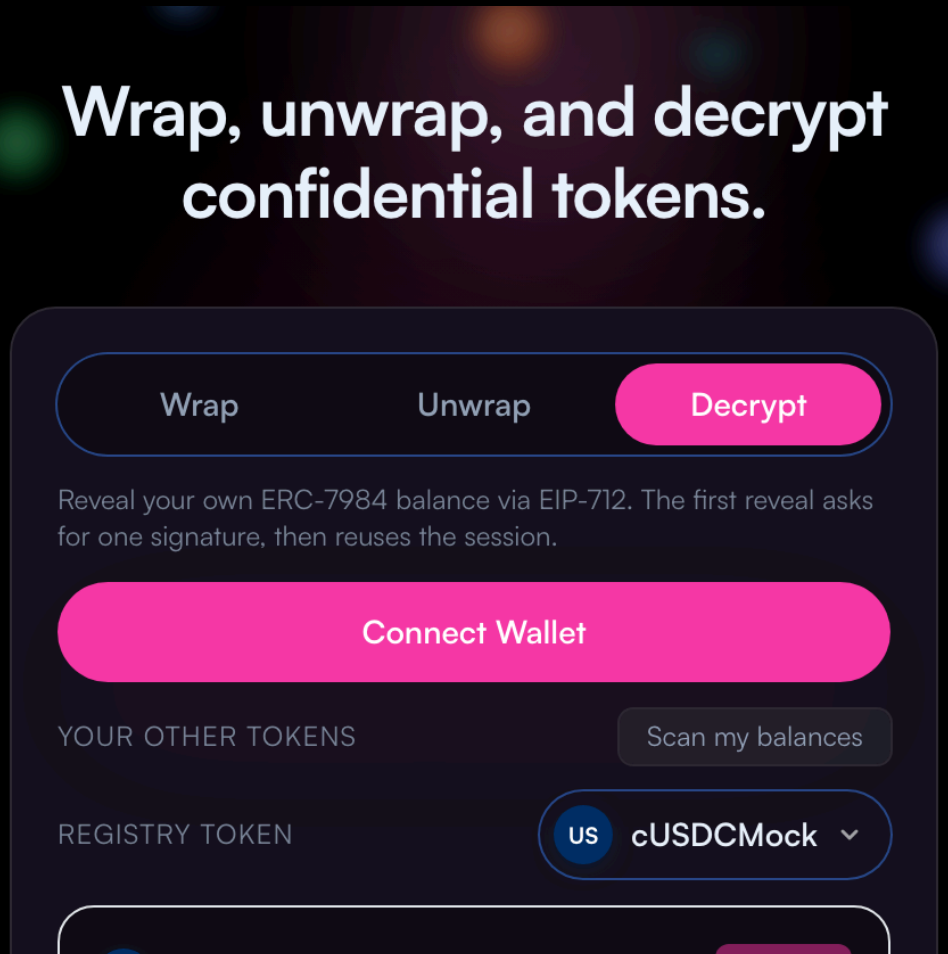
3 **Wrap tx → confidential ERC-7984**

Pre-flight guards catch missing allowance + insufficient balance before you spend gas. Wrapped **1:1** into cUSDCMock — only the ciphertext handle lands on-chain.

Wrap, unwrap, and decrypt confidential tokens.



DECRYPT *the balance*



The freshly-minted ERC-7984 balance is revealed through the **EIP-712 user-decryption** flow.

Sign once → session cached → cleartext balance

Later reveals reuse the session — no repeat prompt. The public only ever saw a ciphertext handle.

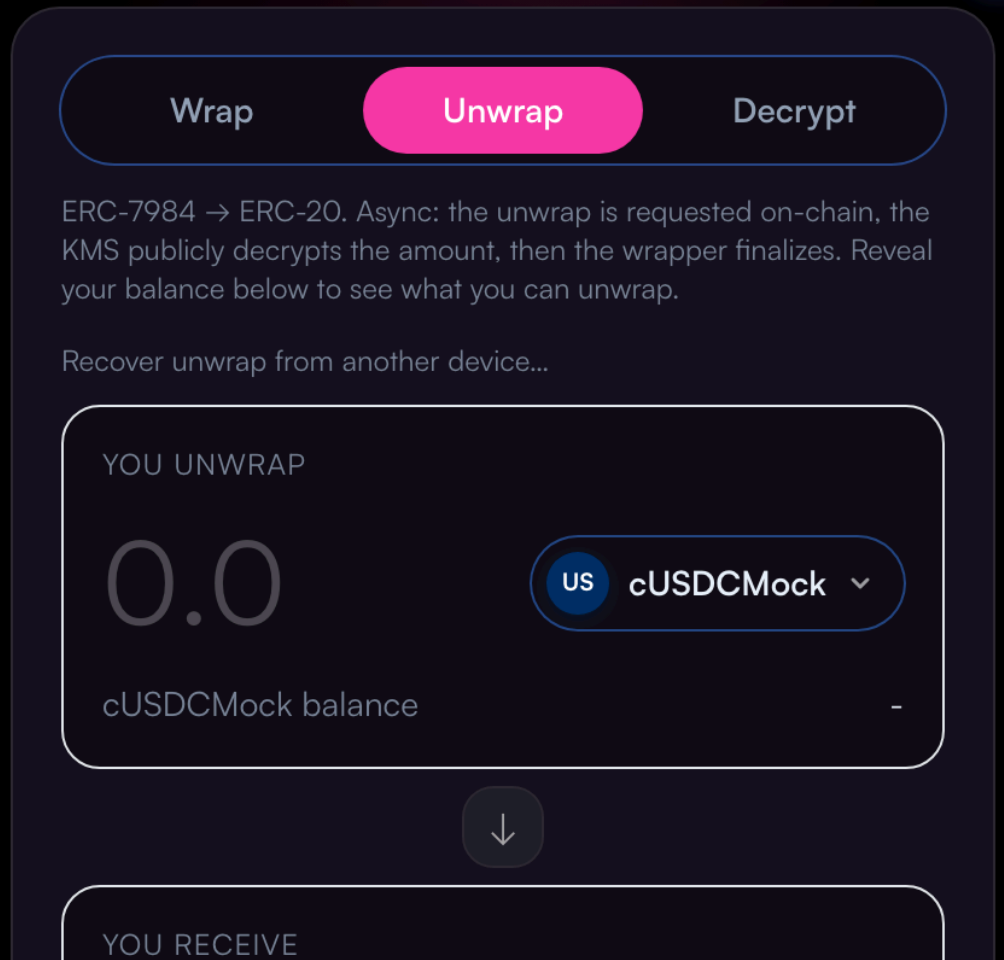
CORRECTNESS

UNWRAP back to ERC-

request → KMS public-decrypt → finalize

A three-step async flow: the plaintext amount is released by Zama's KMS before the contract returns the underlying ERC-20.

Every step is checkpointed to IndexedDB — **resumable**, even across devices, if you refresh mid-flow.



CORRECTNESS · EIP-712

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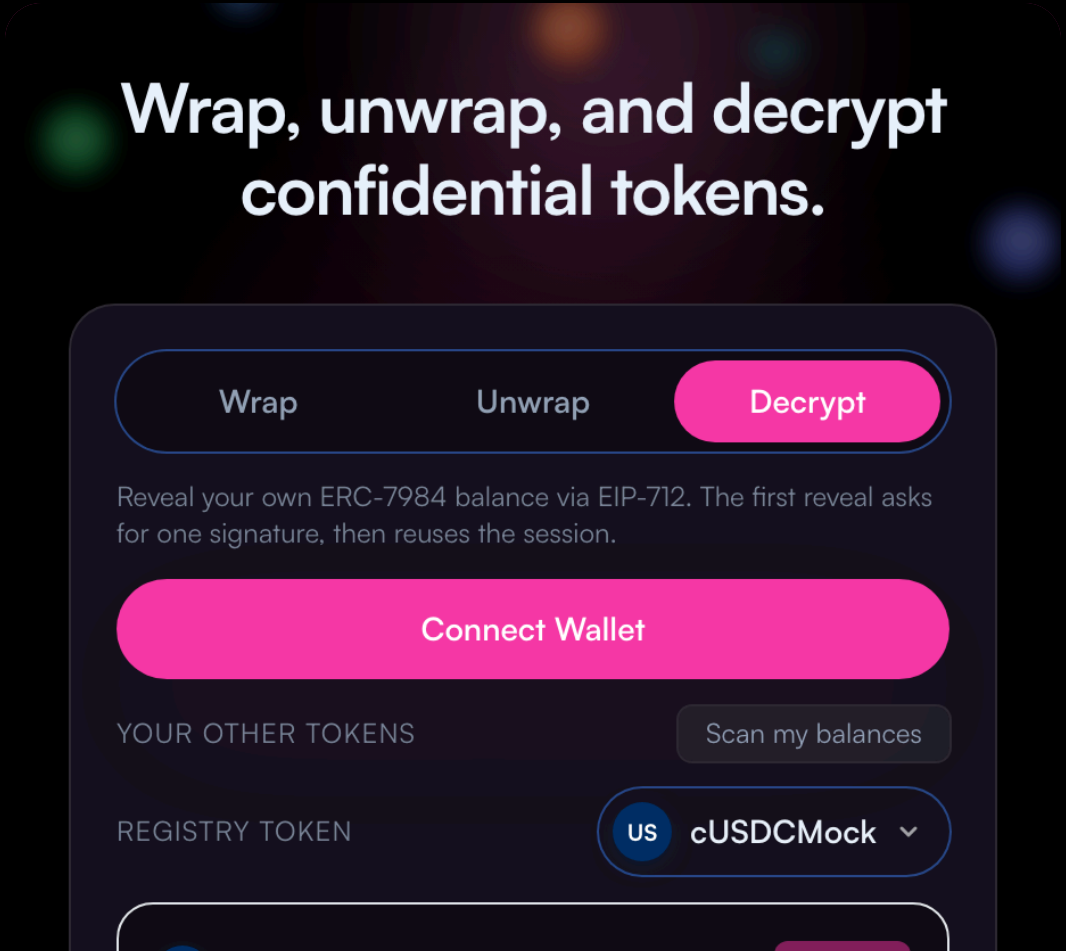
DECRYPT *any ERC-7984*



Paste **any** ERC-7984 address — a token that was never registered here — and reveal *your* balance the same way.

paste 0x... → is-confidential? → reveal

Auto-detect also scans the wallet for non-zero confidential holdings. Decryption isn't limited to the

EXTENSIBILITY

ADD A *new pair*

Two documented paths — on-chain always wins over the local overlay via dedup.

A • On-chain registration

```
# picked up automatically as source: onchain
PRIVATE_KEY=0x… \
ERC20_ADDRESS=0x… \
WRAPPER_ADDRESS=0x… \
  bash scripts/register-pair.sh
```

Documented in the README with this exact example.

B • Local overlay — `config/pairs.ts`

```
CUSTOM_PAIRS.push({
  chainId: sepolia.id,
  erc20Address: "0xYourErc20…",
  confidentialTokenAddress: "0xWrapper…",
  underlying:   { name:"My Token", symbol:"MTK", deci
  confidential: { name:"Conf MTK", symbol:"cMTK", dec
})
```

JUDGED *on*

01 **Coverage**

All 8 official cTokenMocks surfaced from on-chain; dual-chain Sepolia + Mainnet browse.

02 **Correctness**

Wrap, unwrap, and EIP-712 user-decrypt produce the expected on-chain results.

03 **Extensibility**

Two documented add-pair paths — on-chain script + local config — with a real example.

04 **UX**

Approvals, network switching, resumable unwrap, and every error humanized — never a raw revert.

05 **Code quality**

Typed end-to-end, single error funnel, no debug residue, SSR-safe readiness gates.

06 **Production-readiness**

Live on Sepolia + Mainnet, public-RPC fallback, Mainnet writes behind an explicit confirm.

Wrapline

Wrapline — confidential tokens, made usable.

THANK you

wrapline.vercel.app

Not affiliated with Zama. Built on the public Zama Protocol.